DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 9

SQL developer

Profession 9 of 10 in Development and IT.

What the profession means

A SQL developer writes, tunes and maintains the queries and database objects that applications and reports depend on.

What professionals do every day

Writes queries and views, designs tables and keys, optimises slow queries, writes stored procedures where needed, and reviews changes.

Why businesses hire them, and what problems they solve

Slow or wrong queries slow the whole business. A SQL developer keeps the data layer fast and correct.

- Queries that scan whole tables.
- Duplicate or broken data after joins.
- Changes that break reports.

Skills you need

- SELECT, JOIN, GROUP BY and window functions (see Module C).
- Keys, constraints and indexes.
- Query plans and optimisation.
- Version control for SQL.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
PostgreSQL	Full database server	Free	Hosted plans vary	www.postgresql.org
DBeaver	Database client (free community edition)	Free	Paid plans	dbeaver.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write a query with a join and a grouped total.
Intermediate	Add an index and explain the effect on a large table.
Advanced	Write a view that other reports can use.

Projects by level

- **Beginner:** Reporting views for a shop.
- **Intermediate:** Customer 360 query for a firm.
- **Advanced:** Optimisation of a slow dashboard query.

Client-ready portfolio project

A reporting view with two tested queries and a note on the index choice, for a fictional shop.

Worked example

Worked example. The view computes monthly revenue once; three reports read it, so a fix in one place updates all three.

Assignment with full solution

ASSIGNMENT

Reporting view for monthly revenue

Business problem: A shop's three reports each recalculate monthly revenue differently.

Steps

1. Write one query for monthly revenue.
2. Save it as a view.
3. Point each report at the view.
4. Check the monthly totals match the earlier results.

Expected output: The view definition and a matching result.

SOLUTION (ANSWER KEY)

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month;
```

The monthly results are January 3,700; February 2,100; March 5,300. Each report now reads these from the view, so the numbers match everywhere.

Common mistakes

- Using SELECT * in production.
- Joins that multiply rows silently.
- No testing of changes against known results.
- Indexes added everywhere without reason.

Freelancing opportunities

- Database and reporting work for small firms.
- Query tuning for slow systems.
- Briefs for 'SQL', 'PostgreSQL', 'database query'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- PostgreSQL documentation
- SQLBolt

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.